

Lecture 5

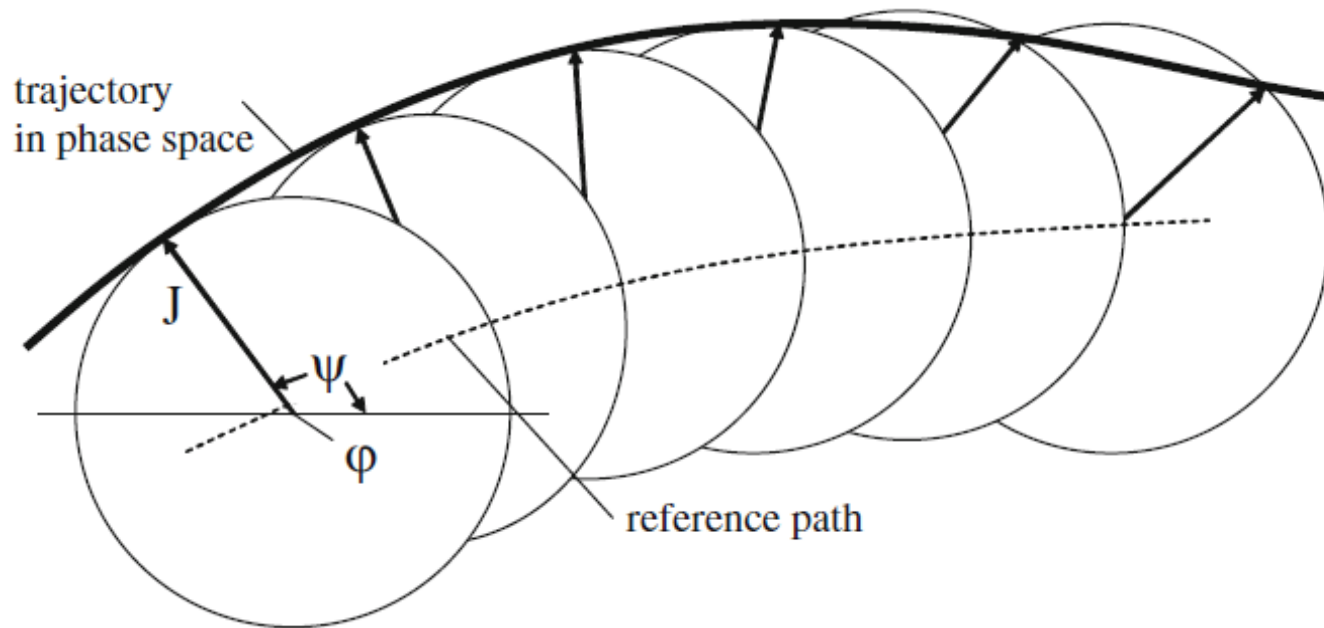
Wednesday 7/15/26

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US Particle Accelerator School
Accelerator Physics – Summer 2026

Action-Angle Coordinates

Trajectories with action-angle coordinates



Wiedemann Fig. 8.7

Courant-Snyder Invariant

Poincaré Map: plotted at fixed s location, evolution of the coordinate of particle turn by turn

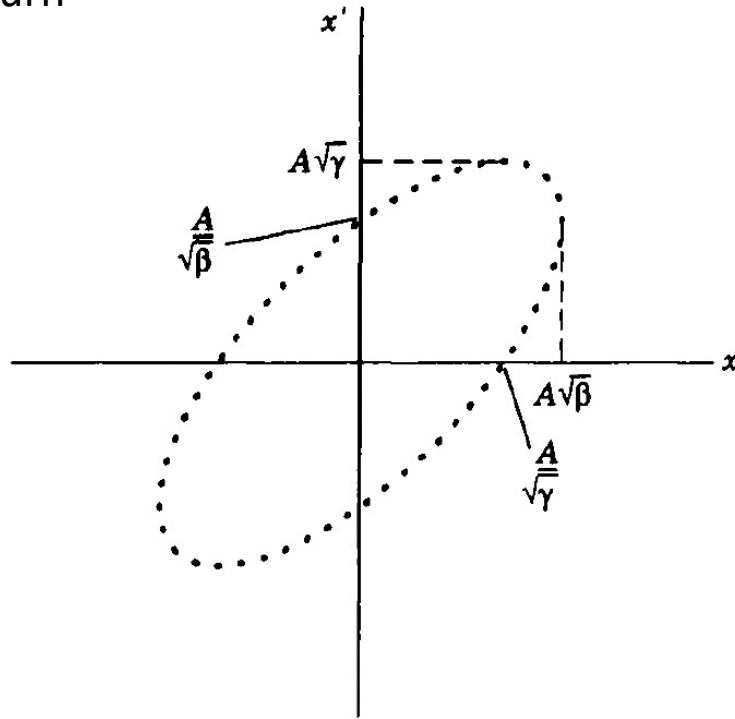


Figure 3.12. Phase space mapping from turn to turn in a circular accelerator.

Ellipse rotates with s , described by CS (Twiss) parameters

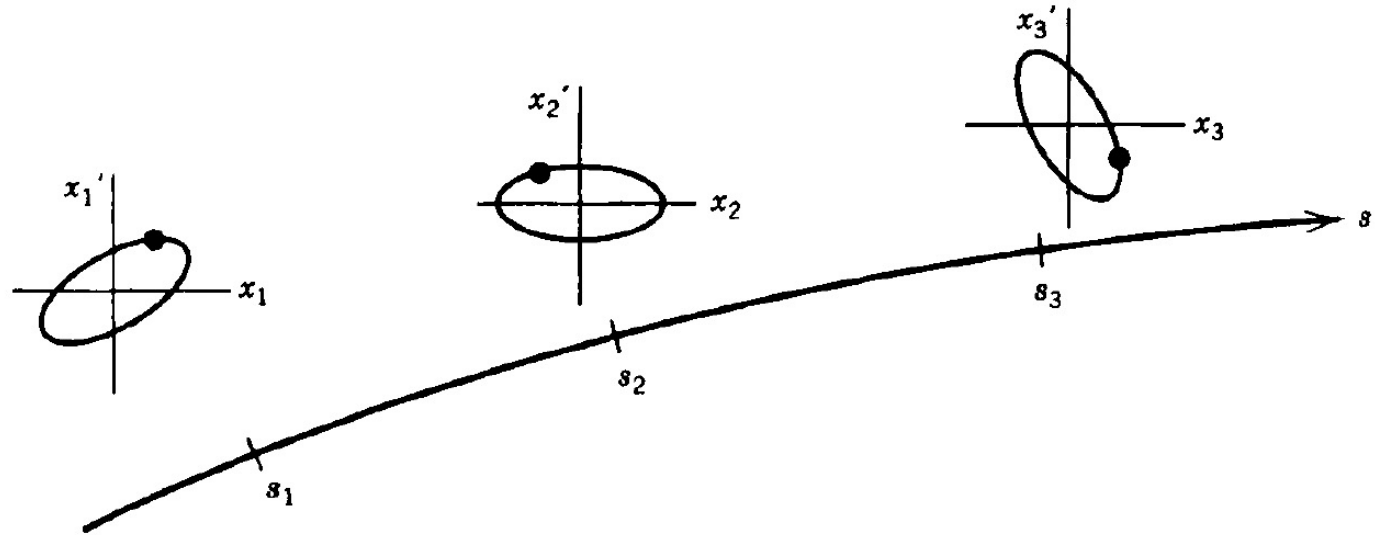
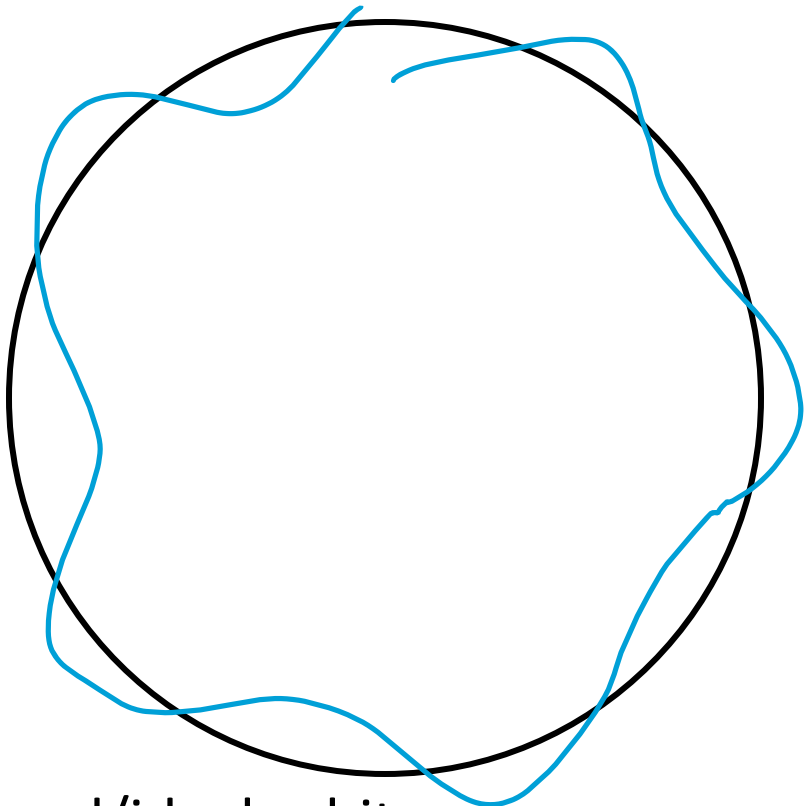


Figure 3.11. Phase space ellipses along the design trajectory.

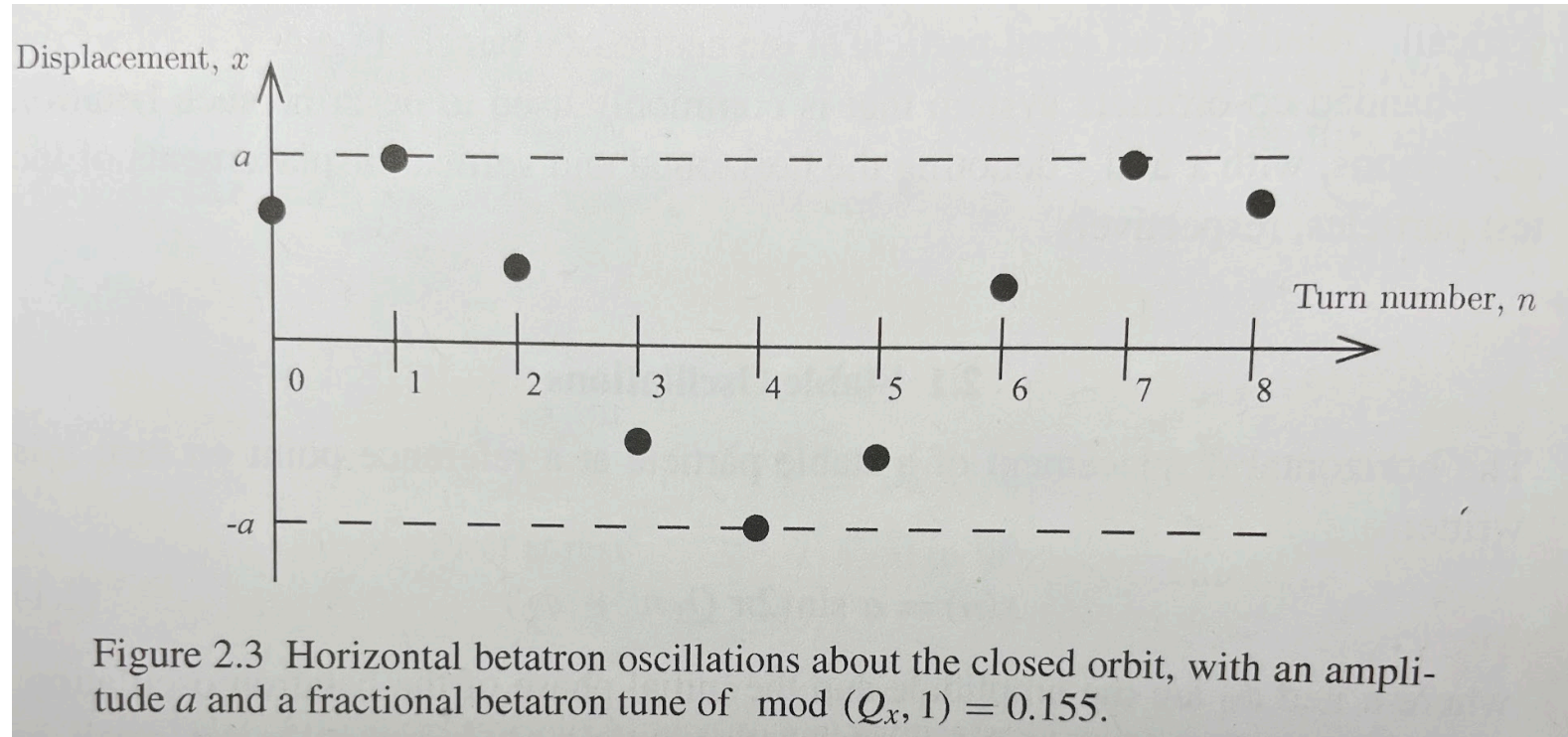
From [Edwards & Syphers Intro to Accelerator Physics](#)

Visualizing betatron oscillations

Betatron oscillation

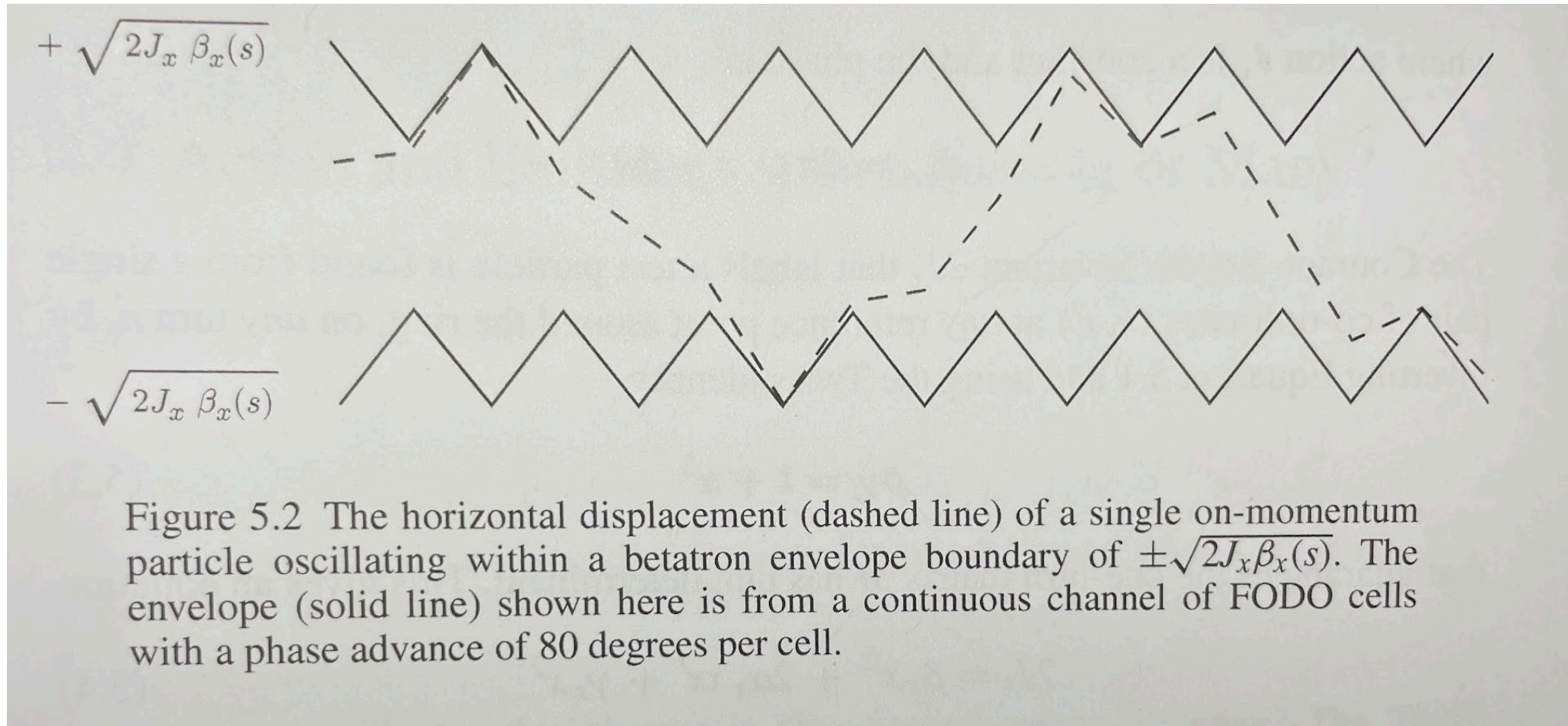


Closed/ideal orbit



S. Peggs and T. Satogata, Fig. 2.3

Visualizing betatron oscillations



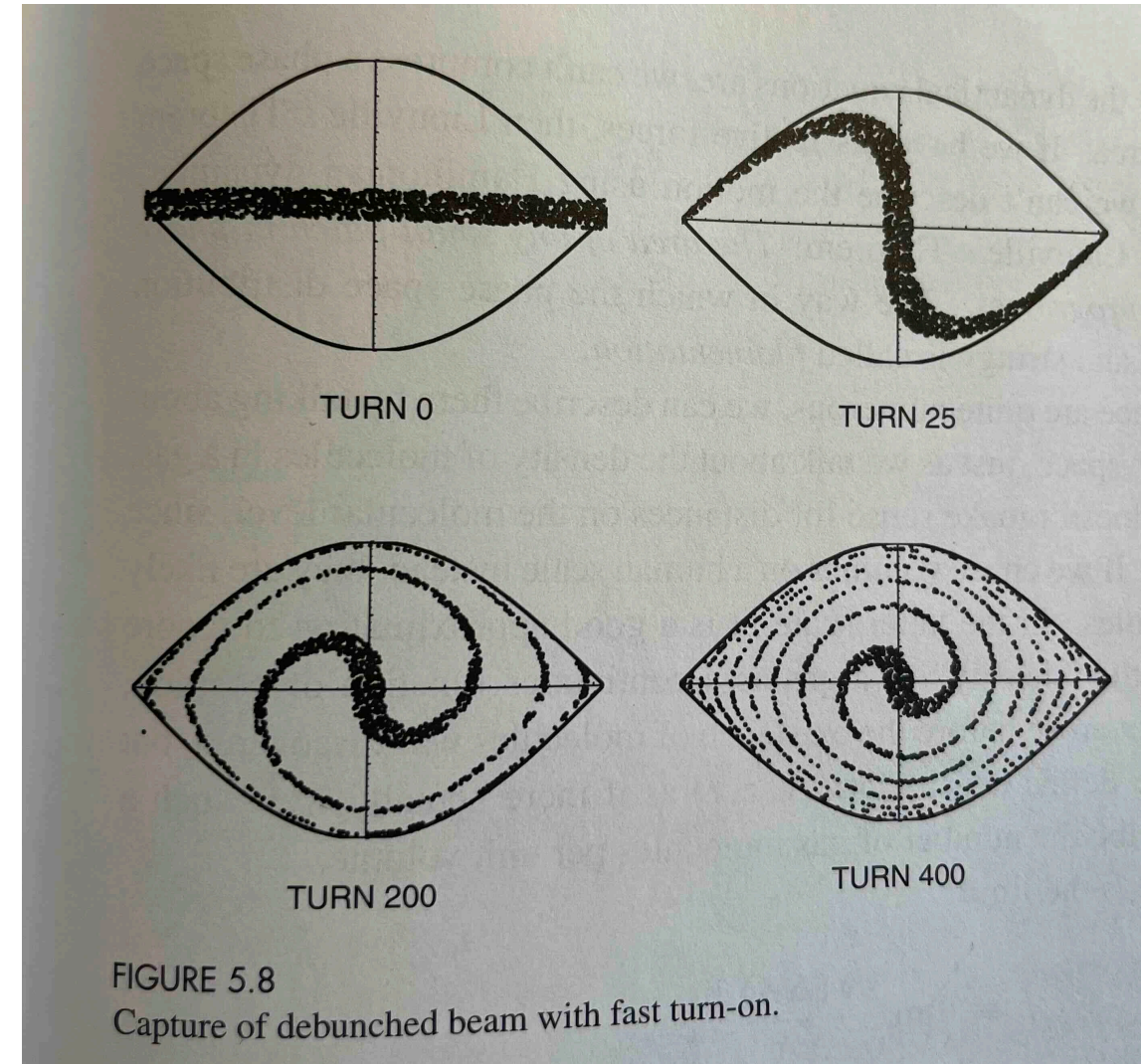
Peggs and Satogata Fig 5.2

Liouville's Theorem

H.W. 8.1.1

Lee A.I.4

- Liouville's Theorem: describes a phase space of a number of identical dynamical systems governed by identical Hamiltonians (in a conservative system – i.e. in the absence of dissipative forces)
- Size of the phase space area is preserved through dynamical evolution. Shape changes, but area is conserved.
- Many points: local density over some small patch is fixed, analogous to incompressible fluid



From Hand and Finch *Analytical Mechanics*, longitudinal phase space evolution of beam in accelerator

CS ellipse in terms of emittance

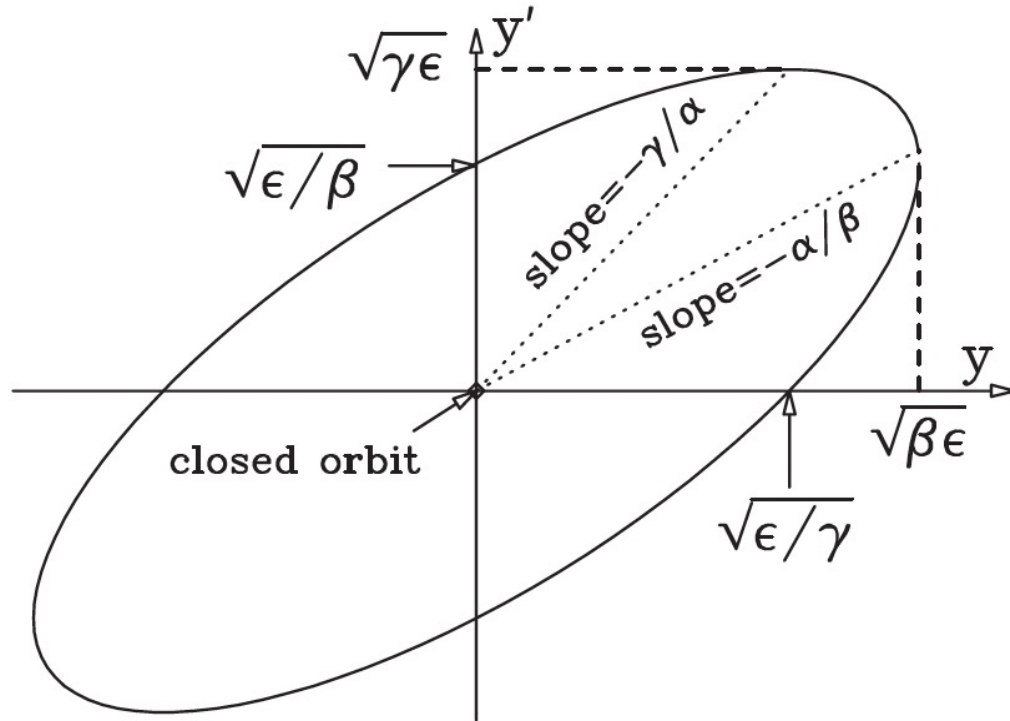
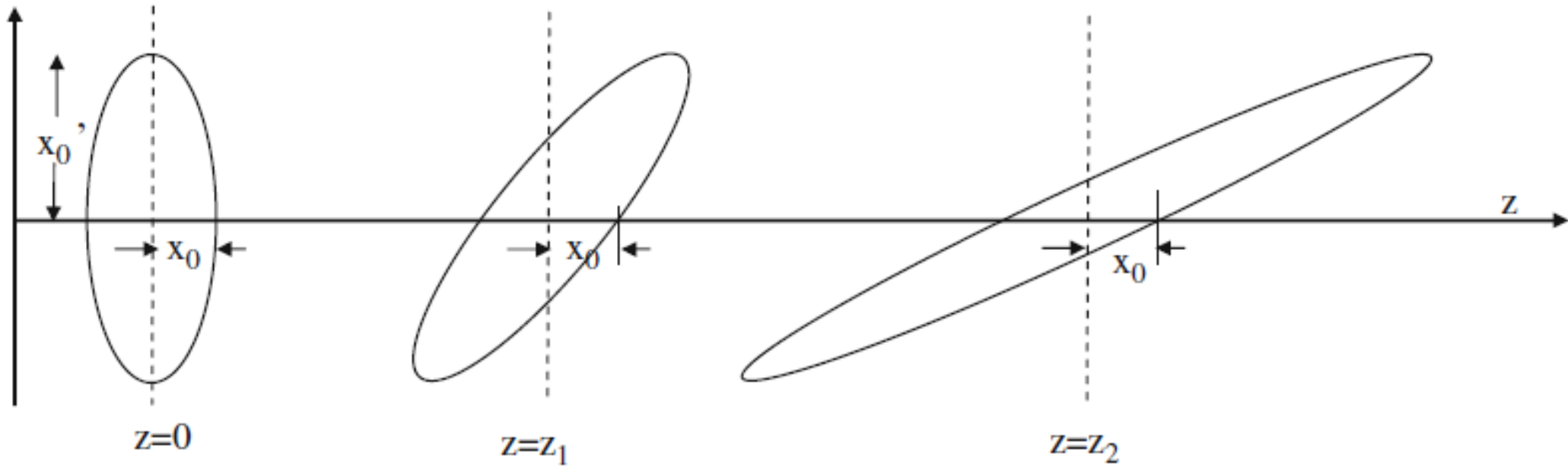


Figure 2.9: The Courant-Snyder invariant ellipse. The area enclosed by the ellipse is equal to $\pi\epsilon$, where ϵ is twice the betatron action; β is the betatron amplitude functions, and $\alpha = -\frac{1}{2}\beta'$, $\gamma = (1 + \alpha^2)/\beta$. The maximum amplitude of betatron motion is $\sqrt{\beta\epsilon}$, and the maximum divergence (angle) of the betatron motion is $\sqrt{\gamma\epsilon}$ (see the footnote in Eq. (2.50) for the convention of the emittance and action). The center of the ellipse is the reference orbit or closed orbit (c.o.).

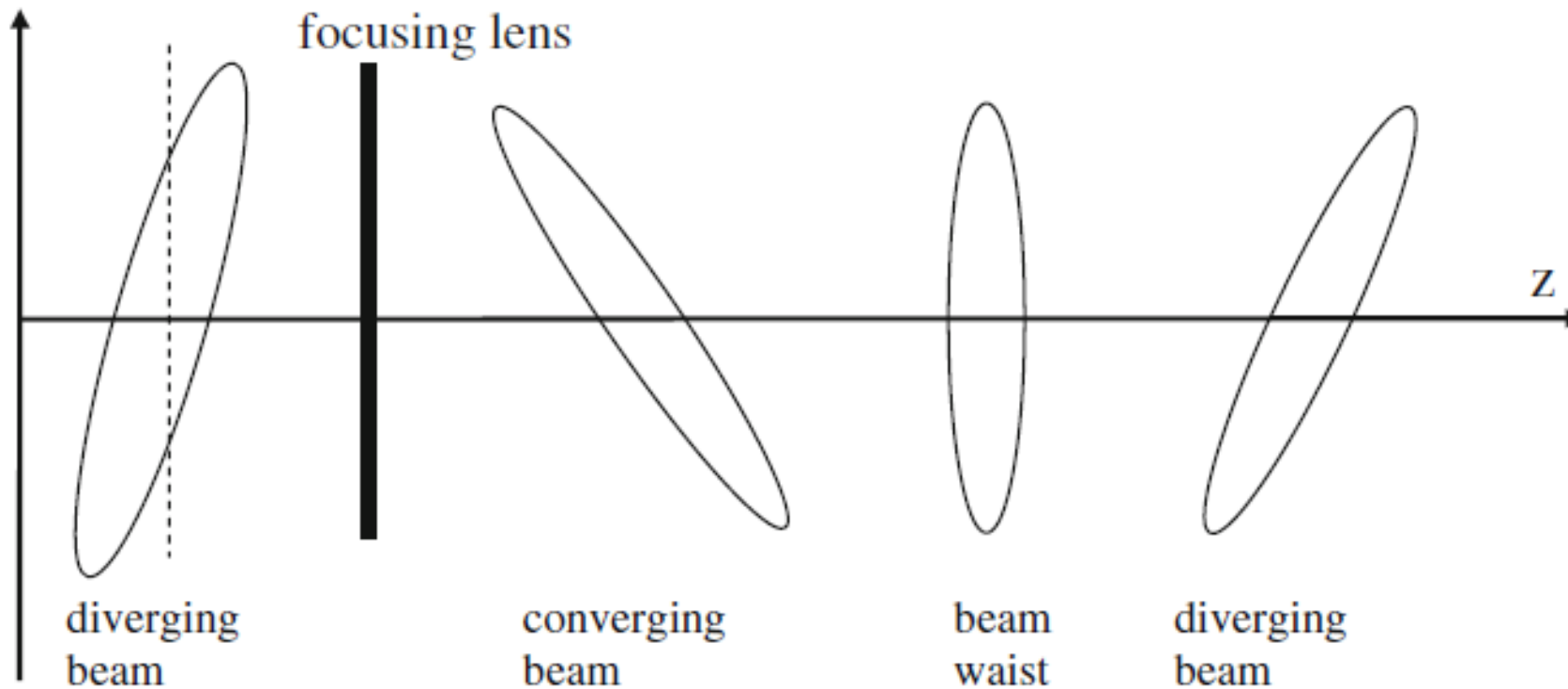
S.Y. Lee Fig 2.9

Evolution through a drift



Wiedemann Fig. 8.3

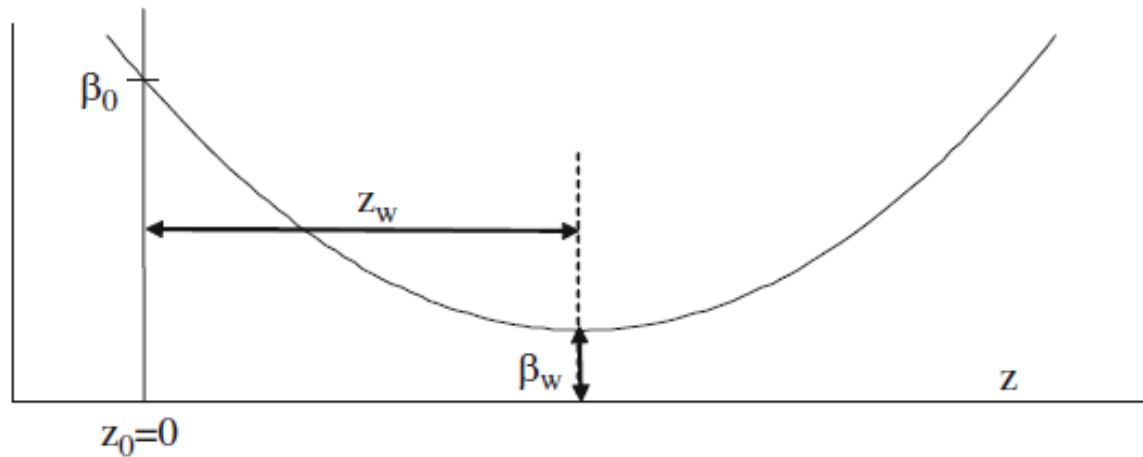
Evolution through focusing quad



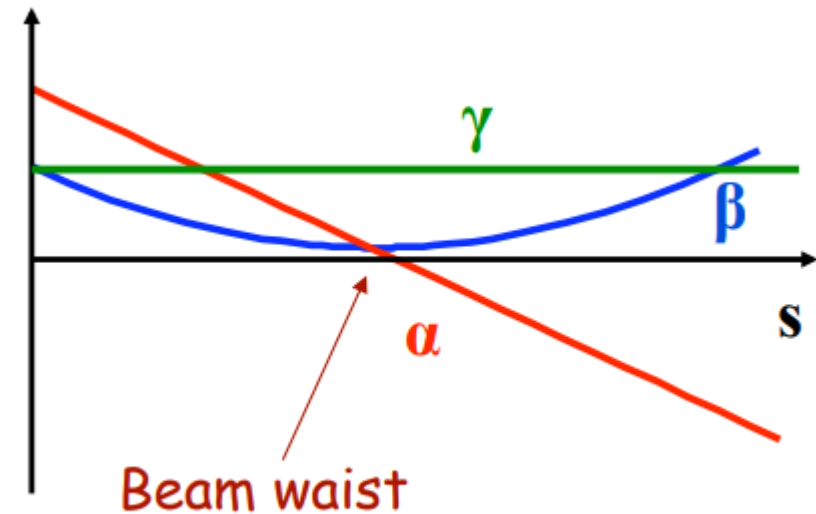
Wiedemann Fig. 8.4

Example of Twiss parameter propagation

Evolution in a drift space near a beam waist (last night's homework)

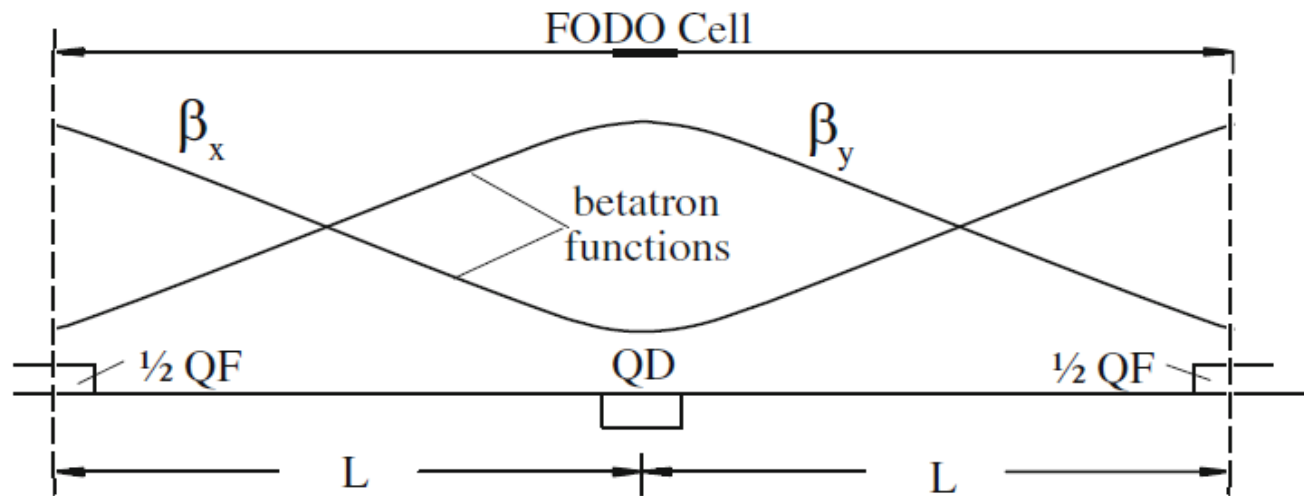


Wiedemann Fig. 8.5

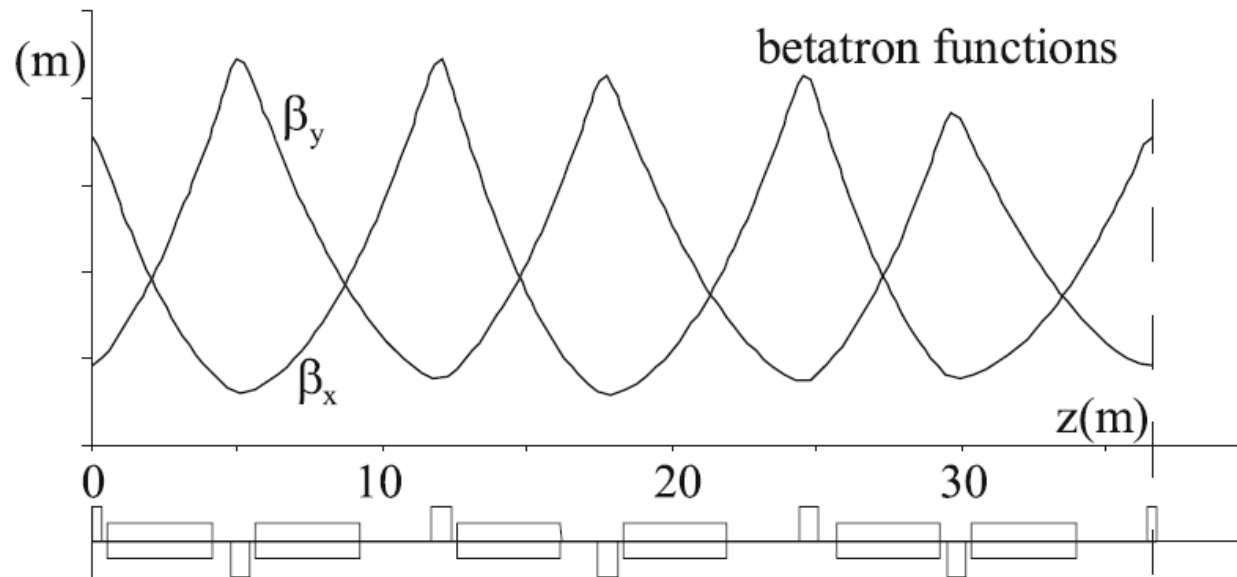


<https://uspas.fnal.gov/materials/09VU/Lecture6.pdf>

Example: FODO cell

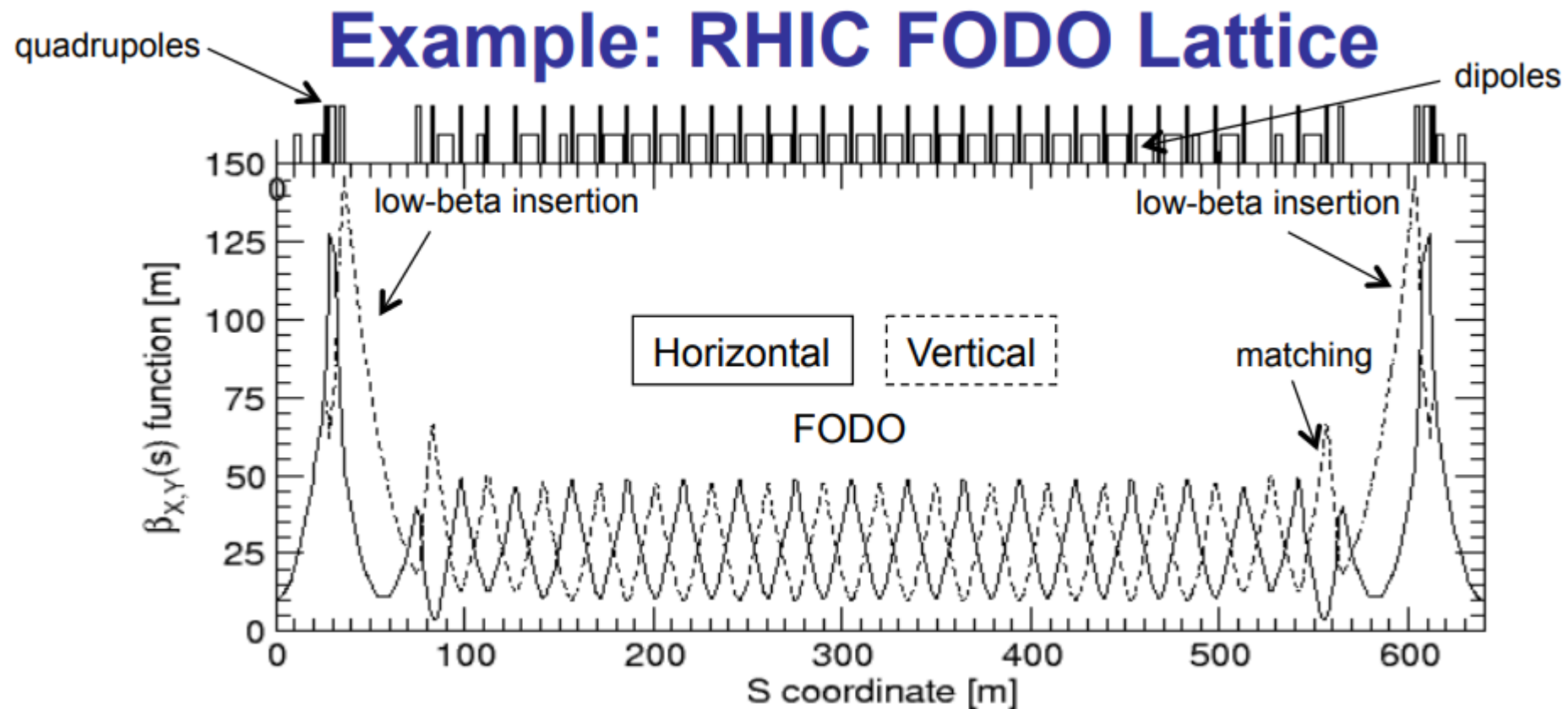


Wiedemann Fig. 10.2, one period



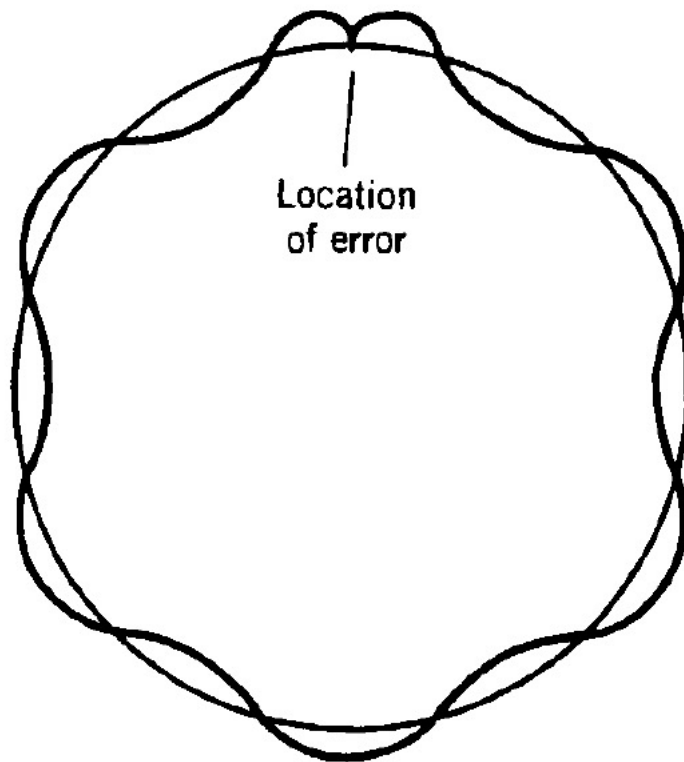
Wiedemann Fig. 10.6, several periods

RHIC example



USPAS winter 2019 materials

Dipole field error



Edwards and Syphers Fig. 3.17

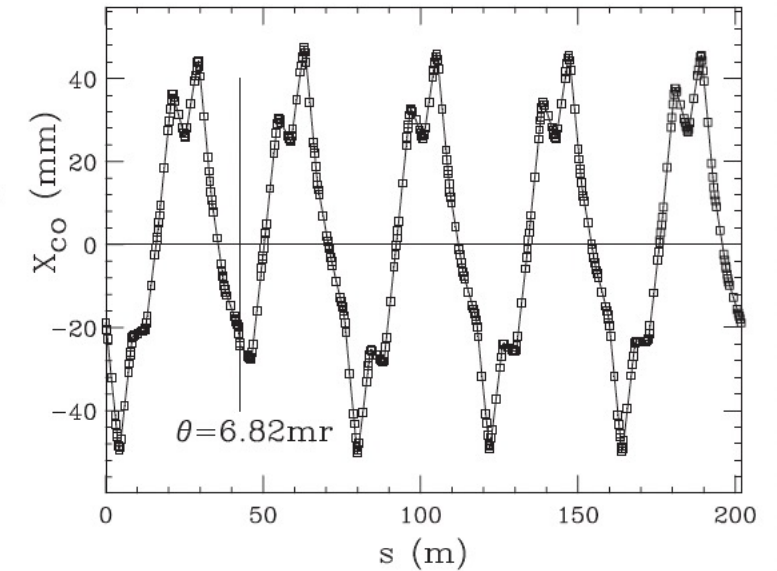
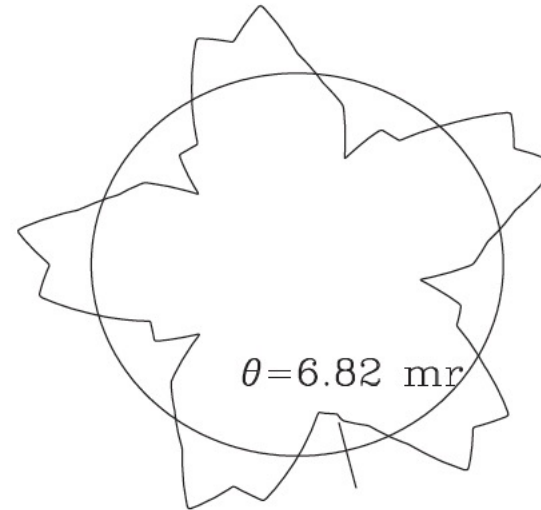


Figure 2.12: Left: schematic plot of the closed-orbit of the AGS booster resulting from a horizontal kicker with kick-angle $\theta = 6.82 \text{ mr}$ at the location marked by a straight line. Since the betatron tune of 4.82 is close to the integer 5, the closed orbit is dominated by the fifth error harmonic. Right: the closed orbit as a function of the longitudinal distance.

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Impact of integer tune

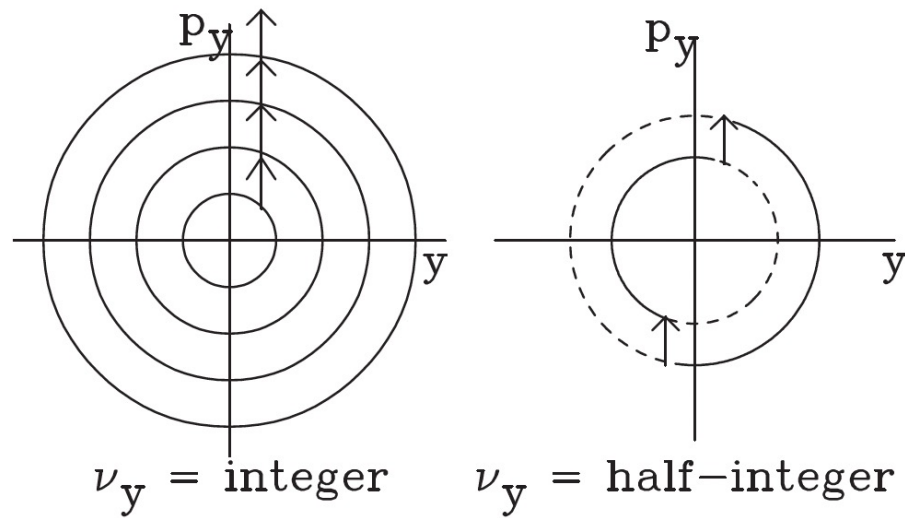


Figure 2.13: Left, a schematic plot of the closed-orbit perturbation due to an error dipole kick when the betatron tune is an integer. Here $\Delta p_y = \beta_y \Delta y' = \beta_y \theta$, where θ is the dipole kick angle and β_y is the betatron amplitude function value at the dipole. Right, a schematic plot of the particle trajectory resulting from a dipole kick when the betatron tune is a half-integer; here the angular kicks from two consecutive orbital revolutions cancel each other.

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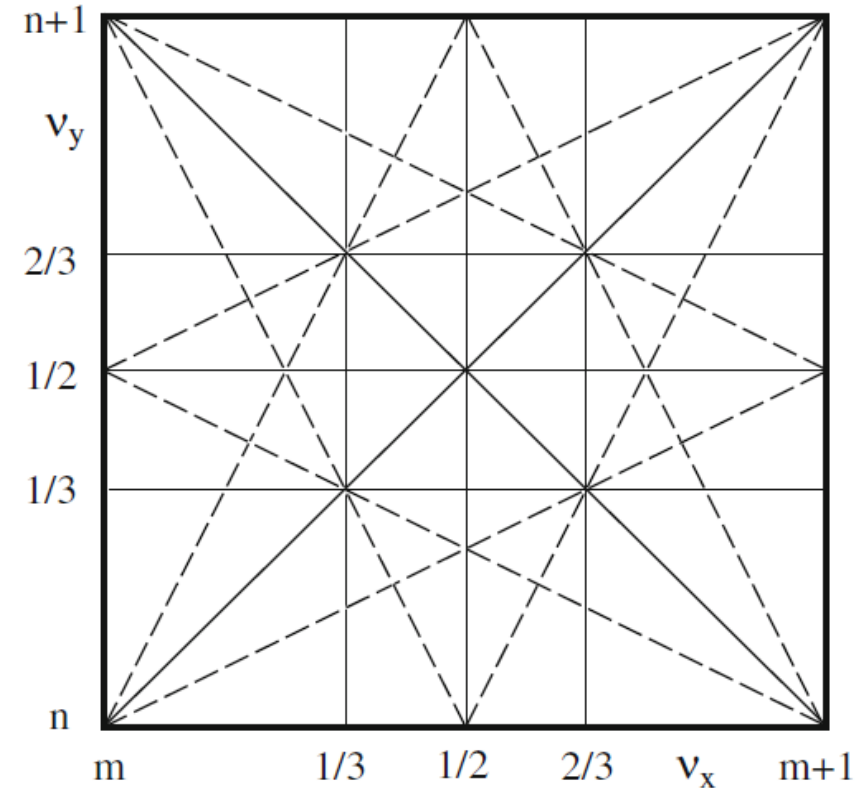
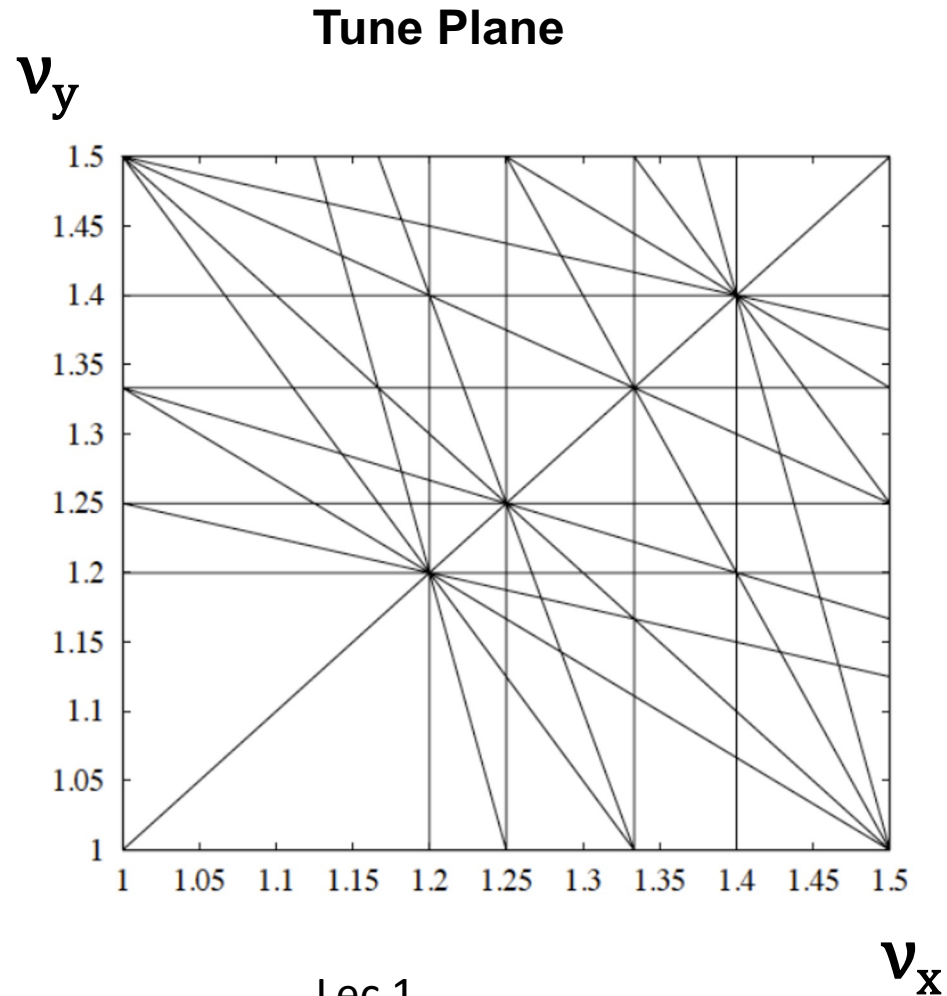
Perturbed one turn transfer matrix

From quadrupole gradient error (SY Lee Sec. 2.III.4)

$$\mathbf{M}(s_1) = \begin{pmatrix} \cos \Phi_0 + \alpha_1 \sin \Phi_0 - \beta_1 k(s_1) ds_1 \sin \Phi_0 & \beta_1 \sin \Phi_0 \\ -\gamma_1 \sin \Phi_0 - [\cos \Phi_0 + \alpha_1 \sin \Phi_0] k(s_1) ds_1 & \cos \Phi_0 - \alpha_1 \sin \Phi_0 \end{pmatrix}$$

Tune map

See Wiedemann Chapter 16 for a good discussion of resonances/resonant instabilities



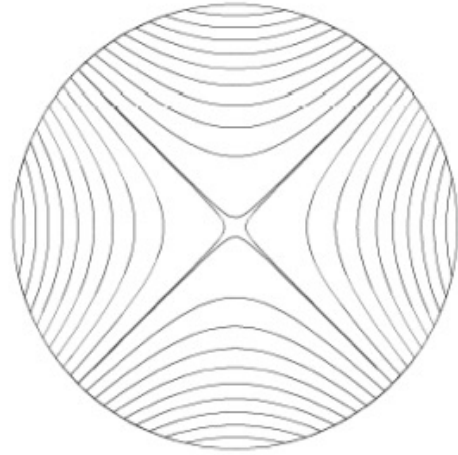
Wiedemann book Fig. 16.2

Multipole expansion

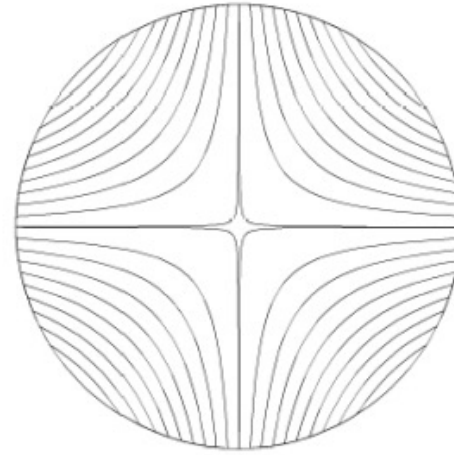
- Textbook: H.W. 6.1, plus some discussion in SY Lee Appendix B.III.8
- Detailed slides from other sources:
 - <https://indico.cern.ch/event/915748/attachments/2037577/3417545/3.pdf>
 - <https://uspas.fnal.gov/materials/14Knoxville/Lecture01.pdf>

Multipole fields

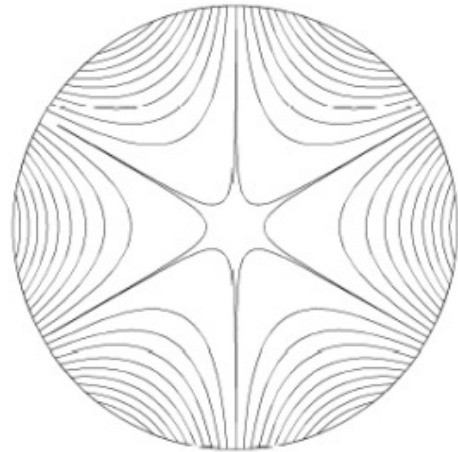
$C_2 = \text{real}$, normal quadrupole



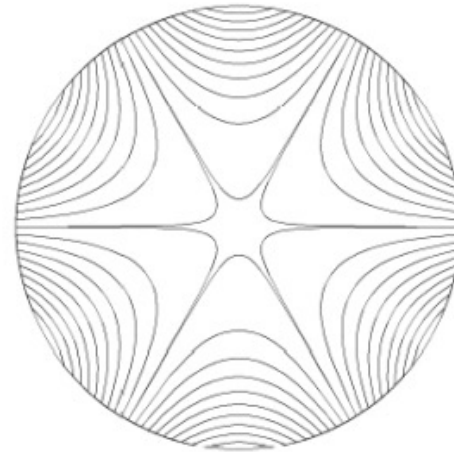
$C_2 = \text{imaginary}$, skew quadrupole



$C_3 = \text{real}$, normal sextupole



$C_3 = \text{imaginary}$, skew sextupole



III.8 2D Magnetic Field in Multipole Expansion

order		Normal multipole ($B_0 b_n$)
n	B_x	B_z
0		$B_0 b_0$
1	$B_0 b_1 z$	$B_0 b_1 x$
2	$B_0 b_2 (2xz)$	$B_0 b_2 (x^2 - z^2)$
3	$B_0 b_3 (3x^2 z - z^3)$	$B_0 b_3 (x^3 - 3xz^2)$
4	$B_0 b_4 (x^3 z - xz^3)$	$B_0 b_4 (x^4 - 6x^2 z^2 + z^4)$
5	$B_0 b_5 (5x^4 z - 10x^3 z^2 + 5xz^4)$	$B_0 b_5 (x^5 - 10x^3 z^2 + 5xz^4)$

order		Skew multipole ($B_0 a_n$)
n	B_x	B_z
0	$B_0 a_0$	
1	$B_0 a_1 x$	$-B_0 a_1 z$
2	$B_0 a_2 (x^2 - z^2)$	$-B_0 a_2 (2xz)$
3	$B_0 a_3 (x^3 - 3xz^2)$	$-B_0 a_3 (3x^2 z - z^3)$
4	$B_0 a_4 (x^4 - 6x^2 z^2 + z^4)$	$-B_0 a_4 (x^3 z - xz^3)$
5	$B_0 a_5 (x^5 - 10x^3 z^2 + 5xz^4)$	$-B_0 a_5 (5x^4 z - 10x^3 z^2 + 5xz^4)$